MOXIFLOXACIN

Most QT prolongation of FQs

No renal adjustment

CHEMICAL NAME 1-cyclopropyl-6-fluoro-8-methoxy-7-[(4aS,7aS)-octahydro-6H-pyrrolo[3,4-b]pyridin-6-yl]-4-oxo-1,4-dihydroquinoline-3-carboxylic acid (hydrochloride)

CLASS Respiratory fluoroquinolone (fourth generation)

TRADE NAMES Avelox · Vigamox (eye drops) · Moxiflox

THERAPEUTIC USES / INDICATIONS

- Community-acquired pneumonia, acute bacterial sinusitis, COPD exacerbation (second-line).
- Complicated skin and intra-abdominal infections (anaerobic cover).
- MDR tuberculosis.
- Eye drops: bacterial conjunctivitis.

ADMINISTRATION

- **PO** 400 mg once daily; separate from antacids, iron, zinc by 4–8 h.
- **IV** infusion over 60 min.
- No renal dose adjustment needed.

DOSAGES

- **Adult:** 400 mg once daily (PO or IV).
- Duration 5–14 days depending on infection.
- **Eye drops:** 1 drop 3 times daily for 7 days.

PHARMACOKINETICS

- **Peak:** 0.5–4 h. Bioavailability ~90%.
- Metabolised in liver (glucuronidation, sulfation — not CYP).
- Excreted in urine and faeces. **t½:** 12 h.

PHARMACODYNAMICS

- Inhibits **DNA gyrase and topoisomerase IV** → bactericidal.
- Good Gram-positive, atypical and anaerobic activity; poor *Pseudomonas* cover.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, diarrhoea.
- Dizziness, headache.
- Taste disturbance.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation (greatest in class).
- Severe hepatotoxicity (rare).
- Tendon rupture; neuropathy; CNS effects.
- Dysglycaemia; *C. difficile*.

CONTRAINDICATIONS & CAUTIONS

- Long QT, bradycardia, heart failure with reduced EF, uncorrected hypokalaemia.
- Severe hepatic impairment.
- Children, pregnancy, breastfeeding; previous FQ tendon injury.

KEY INTERACTIONS

- **QT drugs** (amiodarone, antipsychotics, macrolides): avoid.
- **Antacids, minerals:** ↓ absorption.
- **Warfarin:** ↑ INR.

NURSING MANAGEMENT

- Baseline ECG and electrolytes in at-risk patients.
- Monitor LFTs; watch for jaundice.
- Assess tendon pain and neurological symptoms.
- Monitor glucose in diabetics.
- Monitor clinical response.

PATIENT EDUCATION

- Report palpitations, fainting, yellow skin, tendon pain.
- Take once daily at the same time.
- Avoid antacids or iron near the dose.

KEY POINTS

- Respiratory FQ with anaerobic cover.
- 400 mg once daily; no renal adjustment.
- Highest QT risk; no Pseudomonas.
- Hepatotoxicity possible.

SOURCES CHECKED **[174]** Moxifloxacin 0.5 % w/v eye drops, solution (UK SmPC) **[175]** Moxifloxacin, oral (US label) **[176]** Moxifloxacin, oral (US label, Bryant Ranch Prepack)

Full references p. 432–457